DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM
August 2017
ALGEBRA (Ph.D. Version)

Instructions to the student

a. Answer all six questions; each will be assigned a grade from 0 to 10.

b. Use a different booklet for each question. Write the problem number and your **code number** (**not** your name) on the outside of the booklet.

c. Keep scratch work on separate pages in the same booklet.

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1. Let G be a finite group and let p be the smallest prime dividing $|G|$. Let S be a Sylow p-subgroup of G. Suppose that S is cyclic of order p^n for some $n \geq 1$ and is a normal subgroup of G. Show that S is in the center of G.

2. Let A be an $n \times n$ matrix with entries in a field K. Define a bilinear pairing on K^n by

$$\langle v, w \rangle = (v^T)Aw,$$

where $v, w \in K^n$ are column vectors, and v^T is the row vector that is the transpose of v.
We say that the pairing is non-degenerate if $\langle v, w \rangle = 0$ for all w implies that $v = 0$, and $\langle v, w \rangle = 0$ for all v implies that $w = 0$.
Show that the pairing is non-degenerate if and only if $\det(A) \neq 0$.

3. Let R be an integral domain and let F be its field of fractions. Assume that R has a unique maximal ideal M.
(a) Show that $M = \{r \in R \mid r = 0 \text{ or } 1/r \notin R\}$.
(b) Suppose that R has the property that, for each $0 \neq r \in F$, at least one of r and $1/r$ is in R. Show that every finitely generated ideal of R is principal.

4. Let R be a PID and let M be a finitely generated R-module. Let F be a field containing R. Show that $\mathrm{Hom}_R(M, F)$ (that is, R-module homomorphisms from M to F) and $M \otimes_R F$ have the same dimension as vector spaces over F.

5. Let $n \geq 5$ and let A_n be the group of even permutations of n objects. You may assume the fact that A_n has no normal subgroups except 1 and A_n.
(a) Show that there is no subgroup H of A_n with $1 < [A_n : H] < n$.
(b) Let $f(x) \in \mathbb{Q}[x]$ be an irreducible polynomial of degree $n \geq 5$ and suppose that the splitting field of $f(x)$ has Galois group A_n. Let α be a root of $f(x)$ and let $K = \mathbb{Q}(\alpha)$. Show that if F is a field with $\mathbb{Q} \subseteq F \subseteq K$, then $F = \mathbb{Q}$ or $F = K$.

6. Let G be a finite group and let $\rho : G \to \mathrm{GL}_n(\mathbb{C})$ be a representation of G. Define $\tilde{\rho} : G \times (\mathbb{Z}/2\mathbb{Z}) \to \mathrm{GL}_{2n}(\mathbb{C})$ by

$$\tilde{\rho}(g, 0) = \begin{pmatrix} \rho(g) & 0 \\ 0 & \rho(g) \end{pmatrix}, \qquad \tilde{\rho}(g, 1) = \begin{pmatrix} 0 & \rho(g) \\ \rho(g) & 0 \end{pmatrix}.$$

(a) Show that $\tilde{\rho}$ is a representation of $G \times \mathbb{Z}/2\mathbb{Z}$.
(b) Show that the number of times that the trivial representation of G occurs in ρ equals the number of times that the trivial representation of $G \times \mathbb{Z}/2\mathbb{Z}$ occurs in $\tilde{\rho}$.